

Gaussian Posterior Selection on Noncompact Fibres with Uniformly Separated Roots

YuGuo Li

Abstract

We prove a small-noise Gaussian posterior theorem for the nonlinear family $\langle \langle \text{MATH_D_0} \rangle \rangle$ where $\langle \langle \text{MATH_I_1} \rangle \rangle$ is a $\langle \langle \text{MATH_I_2} \rangle \rangle$ function whose zero set is uniformly separated, whose zeros are uniformly simple, and whose residual has a positive gap outside fixed neighbourhoods of the zeros; no periodicity hypothesis is imposed. Its zero fibre combines a noncompact codimension-one sheet with a countable family of codimension-two points. The main theorem identifies the limiting *ambient-space probability measure*, not merely an unnormalised Laplace coefficient: the posterior selects the sheet and has density proportional to $\langle \langle \text{MATH_I_3} \rangle \rangle$ there. Physical geometric allocation into disjoint cells, Gaussian summability of the countable family, and the explicit complement estimate are derived from the profile hypotheses and the prior, rather than inserted as an atlas summability assumption. A smooth escaping example shows that local regularity alone cannot replace global control.

Keywords: $\setminus \{ \}$ Gaussian posterior; Laplace asymptotics; noncompact fibre; coarea weight; bounded-Lipschitz convergence.

2020 Mathematics Subject Classification: 41A60; 60B10.

1 Gaussian Posterior Selection on Noncompact Fibres with Uniformly Separated Roots

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1.1 Abstract

We prove a small-noise Gaussian posterior theorem for the nonlinear family $\langle \langle \text{MATH_D_0} \rangle \rangle$ where $\langle \langle \text{MATH_I_55} \rangle \rangle$ is a $\langle \langle \text{MATH_I_56} \rangle \rangle$ function whose zero set is uniformly separated, whose zeros are uniformly simple, and whose residual has a positive gap outside fixed neighbourhoods of the zeros; no periodicity hypothesis is imposed. Its zero fibre combines a noncompact codimension-one sheet with a countable family of codimension-two points. The main theorem identifies the limiting *ambient-space probability measure*, not merely an unnormalised Laplace coefficient: the posterior selects the sheet and has density proportional to $\langle \langle \text{MATH_I_57} \rangle \rangle$ there. Physical geometric allocation into disjoint cells, Gaussian summability of the countable family, and the explicit complement estimate are derived from the profile hypotheses and the prior, rather than inserted as an atlas summability assumption. A smooth escaping example shows that local regularity alone cannot replace global control.

Keywords: \{\} Gaussian posterior; Laplace asymptotics; noncompact fibre; coarea weight; bounded-Lipschitz convergence.

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2 Introduction

Gaussian residual posteriors arise when an observation $\ll\text{MATH_I_58}\gg$ is imposed with noise level $\ll\text{MATH_I_59}\gg$; see the Bayesian inverse-problem framework of Stuart [Stuart, 2010]. For a regular compact constraint set, the limiting surface law carries the familiar normal Jacobian (or Fixman/coarea) factor. Geometric Laplace expansions already provide a strong local benchmark [Léger–Vialard, 2023]. That local statement is not the contribution of the present work. The issue addressed here is global: a smooth noncompact fibre can contain infinitely many locally regular pieces, and local Laplace calculations alone do not show that their contributions form a tight normalized posterior.

We isolate a nonlinear family in which this global question has a checkable sharp answer. The factor $\ll\text{MATH_I_60}\gg$ is the minimal residual constraint that creates both a noncompact branch $\ll\text{MATH_I_61}\gg$ and a root-indexed isolated branch $\ll\text{MATH_I_62}\gg$; the family therefore serves as an analyzable testbed for posterior competition between normal codimensions. Figure 1 records the exact geometry for the concrete nonperiodic profile introduced in Proposition 1; it is included for orientation, not as numerical evidence. Let $\ll\text{MATH_I_63}\gg$ be $\ll\text{MATH_I_64}\gg$, let $\ll\text{MATH_I_65}\gg$ be nonempty, and, for $\ll\text{MATH_I_66}\gg$, write $\ll\text{MATH_I_67}\gg$. Suppose that there are $\ll\text{MATH_I_68}\gg$ with $\ll\text{MATH_I_69}\gg$ and $\ll\text{MATH_I_70}\gg$, such that $\ll\text{MATH_D_1}\gg$. These are F-side hypotheses: they mention neither a chart atlas nor a posterior estimate. The map $\ll\text{MATH_I_71}\gg$ has the exact fibre

The displayed fibre and the physical allocation in the figure are the objects used throughout the proof below.

Thus a codimension-one sheet competes with countably many codimension-two points. The standard Gaussian prior makes this a genuinely noncompact problem. The class is strictly broader than the periodic-function special case: a periodic profile with a nonempty finite set of regular roots per period satisfies these hypotheses, while neither the locations of the roots nor the profile need repeat here.

Our result is intentionally narrower than a theorem for arbitrary noncompact fibres, but not a one-profile calculation. It is a theorem for a periodicity-free, uniformly controlled class. Every global estimate is traceable to uniform root separation, explicit residual coercivity, and Gaussian decay. The Lean supplement verifies the exact fibre and energy identities; the analytic packing, integration, and probability-law argument are given in the text and companion proof files.

3 Verifiable profile hypotheses and their role

The hypotheses are imposed on the residual itself, rather than on a chosen atlas or on a prospective limiting law. They have three distinct roles. Separation bounds the number of roots per unit interval and hence controls the Gaussian root sum. Uniform simplicity provides a common quadratic normal-energy bound in every isolated cell. The exterior gap rules out mass away from the sheet and the selected root cells at an exponential small-noise scale. None of these three conclusions is assumed separately.

∴ {#lem:periodic-admissible.lemma} **Lemma 1** (Periodic profiles are admissible). *Let $\ll\text{MATH_I_79}\gg$ be $\ll\text{MATH_I_80}\gg$ -periodic. If $\ll\text{MATH_I_81}\gg$ has a finite nonempty set of zeros in $\ll\text{MATH_I_82}\gg$*

and every such zero is simple, then $\langle \text{MATH_I_83} \rangle$ satisfies the three F-side hypotheses for some $\langle \text{MATH_I_84} \rangle$. $\therefore$

$\therefore$ proof *Proof*. Distinct zeros in one period have a positive minimum separation; periodicity extends this to all of $\langle \text{MATH_I_85} \rangle$, yielding $\langle \text{MATH_I_86} \rangle$. Since $\langle \text{MATH_I_87} \rangle$ is nonzero at each of finitely many zeros, continuity gives a common neighbourhood radius and a common lower bound $\langle \text{MATH_I_88} \rangle$. Choose $\langle \text{MATH_I_89} \rangle$ smaller than this radius and than $\langle \text{MATH_I_90} \rangle$. The complement in $\langle \text{MATH_I_91} \rangle$ of the $\langle \text{MATH_I_92} \rangle$ -neighbourhoods of the finitely many zeros is compact and contains no zero of $\langle \text{MATH_I_93} \rangle$. The positive minimum of $\langle \text{MATH_I_94} \rangle$ there is an admissible $\langle \text{MATH_I_95} \rangle$, and periodicity extends the same gap to the real line. $\therefore$

$\therefore$ remark **Remark 1** (Why periodicity is absent from the theorem). *The preceding lemma is only a convenient verification device. The proof below uses neither a period nor a fundamental domain: it uses only the three displayed F-side conditions. Thus any nonperiodic profile for which those literal conditions can be checked belongs to the same theorem class.* $\therefore$

4 Concrete nonperiodic profiles and the scope of the hypotheses

The periodic verification lemma is useful, but it does not describe the whole class in the theorem. The following elementary family makes the periodicity-free content explicit and also records why the three F-side conditions are natural global hypotheses rather than notation for a local implicit-function argument.

$\therefore$ {#prop:aperiodic-family .proposition} **Proposition 2** (A nonperiodic admissible family). *Let $\langle \text{MATH_I_96} \rangle$ satisfy $\langle \text{MATH_I_97} \rangle$ for every $\langle \text{MATH_I_98} \rangle$, and set $\langle \text{MATH_D_3} \rangle$. Then $\langle \text{MATH_I_99} \rangle$ satisfies the F-side hypotheses. In particular, $\langle \text{MATH_D_4} \rangle$ produces an admissible nonperiodic profile.* $\therefore$

$\therefore$ proof *Proof*. The zero set is $\langle \text{MATH_I_100} \rangle$, hence it is separated by $\langle \text{MATH_I_101} \rangle$. Choose $\langle \text{MATH_I_102} \rangle$. For $\langle \text{MATH_I_103} \rangle$ and $\langle \text{MATH_I_104} \rangle$, the elementary inequality $\langle \text{MATH_I_105} \rangle$ gives $\langle \text{MATH_D_5} \rangle$. If $\langle \text{MATH_I_106} \rangle$, periodicity of the sine factor gives $\langle \text{MATH_I_107} \rangle$, and hence $\langle \text{MATH_D_6} \rangle$. Thus $\langle \text{MATH_I_108} \rangle$ and $\langle \text{MATH_I_109} \rangle$ work. The displayed choice of $\langle \text{MATH_I_110} \rangle$ is smooth, bounded above and below by positive constants, and is not periodic because it has distinct limits at $\langle \text{MATH_I_111} \rangle$ and $\langle \text{MATH_I_112} \rangle$. $\therefore$

This example is deliberately simple. It shows that periodic root placement is not needed, and it isolates the genuinely relevant information: one physical radius which works at every root, plus a residual gap away from all of those root neighbourhoods. Neither condition follows from the statement that each root is individually regular. For example, a sequence of regular roots can have slopes tending to zero, leaving no shared coercivity constant; the escaping example in Section [12](#sec:sharpness){reference-type="ref" reference="sec:sharpness"} exhibits exactly this phenomenon. Likewise, a nonzero residual may become arbitrarily small far from its zero set, so a local chart calculation does not by itself control the posterior mass in the physical complement.

There are two useful ways to read the theorem class. First, it is a class of residuals: its assumptions can be verified before any posterior is formed, and do not contain a summable atlas weight, an enumeration of roots, or a tightness conclusion. Second, it is a class of noncompact fibres with one globally selected codimension-one component and a countable set of competing codimension-two components. The proof uses the standard Gaussian prior to derive a summable weight for the latter components. Replacing the prior by a different tail profile would require rechecking that derived summability step; it is not hidden in the definition of an admissible profile.

5 Posterior and theorem

Let $\llbracket \text{MATH_I_113} \rrbracket$. For $\llbracket \text{MATH_I_114} \rrbracket$, define the actual posterior probability measure $\llbracket \text{MATH_D_7} \rrbracket$. No branch label or surrogate measure occurs in this definition.

Set $\llbracket \text{MATH_D_8} \rrbracket$ and define $\llbracket \text{MATH_I_115} \rrbracket$ by $\llbracket \text{MATH_D_9} \rrbracket$. The integral is finite and positive because $\llbracket \text{MATH_I_116} \rrbracket$.

Let $\llbracket \text{MATH_I_117} \rrbracket$ be the common physical radius in the hypotheses and write $\llbracket \text{MATH_I_118} \rrbracket$ for $\llbracket \text{MATH_I_119} \rrbracket$. Write throughout $\llbracket \text{MATH_D_10} \rrbracket$

$\vdash \{\text{\#thm:periodic-selection .theorem}\}$ **Theorem 1** (Uniformly-separated profile posterior selection). *For every $\llbracket \text{MATH_I_120} \rrbracket$ profile $\llbracket \text{MATH_I_121} \rrbracket$ satisfying the F-side hypotheses above (with no periodicity assumption), the associated cells can be chosen so that $\llbracket \text{MATH_D_11} \rrbracket$ and $\llbracket \text{MATH_D_12} \rrbracket$. Here the implicit constant and the small-noise threshold depend only on $\llbracket \text{MATH_I_122} \rrbracket$. In particular, the countable codimension-two part of the fibre receives vanishing mass. Equivalently, for every bounded Lipschitz $\llbracket \text{MATH_I_123} \rrbracket$, $\llbracket \text{MATH_D_13} \rrbracket$*

$\vdash$

5.1 Proof architecture

The proof separates into four estimates, each at the natural evidence scale $\llbracket \text{MATH_I_124} \rrbracket$. First, the sheet tube has a rescaled ambient integral converging to the coarea-weighted line integral in (4). Second, every root cell has mass $\llbracket \text{MATH_I_125} \rrbracket$, uniformly in $\llbracket \text{MATH_I_126} \rrbracket$, and Gaussian packing makes their sum $\llbracket \text{MATH_I_127} \rrbracket$. Third, the literal exterior gap makes the remaining physical complement exponentially small. Fourth, the positive sheet limit normalizes the preceding estimates and yields both tightness and bounded-Lipschitz convergence. The next two sections prove these statements in this order; recording the scale separation explicitly is essential because the fibre contains strata of different normal codimension.

The four components are stated separately below so that the normalisation argument does not conceal a local or countable-family assumption.

$\vdash \{\text{\#lem:sheet-scaling .lemma}\}$ **Lemma 2** (Sheet-tube scaling). *For every bounded continuous $\llbracket \text{MATH_I_128} \rrbracket$ and $\llbracket \text{MATH_I_129} \rrbracket$, $\llbracket \text{MATH_D_14} \rrbracket$*

$\vdash$

$\vdash \{\text{\#lem:root-cells .lemma}\}$ **Lemma 3** (Countable root-cell contribution). *There is a constant $\llbracket \text{MATH_I_130} \rrbracket$ such that, for every $\llbracket \text{MATH_I_131} \rrbracket$ and every $\llbracket \text{MATH_I_132} \rrbracket$, $\llbracket \text{MATH_D_15} \rrbracket$ Consequently $\llbracket \text{MATH_I_133} \rrbracket$.*

$\vdash$

$\vdash \{\text{\#lem:exterior-suppression .lemma}\}$ **Lemma 4** (Physical-complement suppression). *With $\llbracket \text{MATH_I_134} \rrbracket$, $\llbracket \text{MATH_D_16} \rrbracket$ The same estimate, up to multiplication by $\llbracket \text{MATH_I_135} \rrbracket$, holds with a bounded test function in the integrand.*

$\vdash$

$\vdash \{\text{\#prop:posterior-assembly .proposition}\}$ **Proposition 3** (Normalization, tightness, and posterior assembly). *The three preceding estimates imply $\llbracket \text{MATH_D_17} \rrbracket$ and $\llbracket \text{MATH_I_136} \rrbracket$ weakly, hence in bounded-Lipschitz distance. Moreover $\llbracket \text{MATH_I_137} \rrbracket$ is uniformly tight for every sufficiently small $\llbracket \text{MATH_I_138} \rrbracket$, and $\llbracket \text{MATH_D_18} \rrbracket$*

$\vdash$

6 Geometry supplied by literal profile hypotheses

The exact identities underlying the proof are $\llbracket \text{MATH_D_19} \rrbracket$. For $\llbracket \text{MATH_I_139} \rrbracket$, the first identity is at least $\llbracket \text{MATH_I_140} \rrbracket$. Hence the sheet has a uniform physical normal scale even though its Jacobian weight is not constant.

The following lemma records all global information used later. It is the point at which physical root separation and the literal exterior condition replace a separately assumed countable-atlas

majorant. Set $\langle\langle \text{MATH_D_20} \rangle\rangle$

∴ {#lem:periodic-geometry .lemma} **Lemma 5** (Uniform cells, Gaussian packing, and physical exterior gap). *The shared radius $\langle\langle \text{MATH_I_141} \rangle\rangle$ makes the cells $\langle\langle \text{MATH_I_142} \rangle\rangle$ pairwise disjoint, disjoint from $\langle\langle \text{MATH_I_143} \rangle\rangle$, and $\langle\langle \text{MATH_D_21} \rangle\rangle$. Moreover, $\langle\langle \text{MATH_D_22} \rangle\rangle$. In addition, $\langle\langle \text{MATH_I_144} \rangle\rangle$. ∴*

∴ proof *Proof*. The shared radius satisfies $\langle\langle \text{MATH_I_145} \rangle\rangle$, so the cells are disjoint, and the uniform coercivity is exactly the second F-side hypothesis. Each interval $\langle\langle \text{MATH_I_146} \rangle\rangle$ contains at most $\langle\langle \text{MATH_I_147} \rangle\rangle$ roots. Comparing root locations in that interval with $\langle\langle \text{MATH_I_148} \rangle\rangle$ yields $\langle\langle \text{MATH_D_23} \rangle\rangle$. This derives, rather than assumes, the only countable-family summability used below. For $\langle\langle \text{MATH_I_149} \rangle\rangle$, either $\langle\langle \text{MATH_I_150} \rangle\rangle$, in which case $\langle\langle \text{MATH_I_151} \rangle\rangle$ because the point lies outside $\langle\langle \text{MATH_I_152} \rangle\rangle$, and hence $\langle\langle \text{MATH_I_153} \rangle\rangle$, or $\langle\langle \text{MATH_I_154} \rangle\rangle$. In the latter case $\langle\langle \text{MATH_I_155} \rangle\rangle$, and membership in the displayed complement gives $\langle\langle \text{MATH_I_156} \rangle\rangle$ for every $\langle\langle \text{MATH_I_157} \rangle\rangle$; hence $\langle\langle \text{MATH_I_158} \rangle\rangle$. This proves the displayed exterior bound explicitly and produces a literal physical partition rather than a sum over overlapping charts. ∴

Role of uniform separation. Uniform separation is the geometric input that yields the Gaussian packing bound $\langle\langle \text{MATH_I_159} \rangle\rangle$. It is not sufficient by itself: the uniform local coercivity and the exterior residual gap are the other two ingredients used above. Proposition [6]({#prop:escaping-sharpness}{reference-type="ref" reference="prop:escaping-sharpness"}) shows how global posterior tightness can fail when this uniform residual control is lost, even though its integer zero set remains separated.

7 Normal Jacobians and evidence-scale selection

The limiting density in Theorem [1]({#thm:periodic-selection}{reference-type="ref" reference="thm:periodic-selection"}) is not inserted by a heuristic coarea prescription. Its normal factor can be read directly from the residual. The derivative is $\langle\langle \text{MATH_D_24} \rangle\rangle$. At a point $\langle\langle \text{MATH_I_160} \rangle\rangle$ of the sheet, this has rank one and its nonzero singular value is $\langle\langle \text{MATH_D_25} \rangle\rangle$. Thus the standard normal-coordinate calculation would suggest the density "prior divided by $\langle\langle \text{MATH_I_161} \rangle\rangle$ ". The sheet-tube lemma below proves precisely this suggestion for the actual ambient posterior, including its normalizing constant and the treatment of all competing regions. In particular, the factor $\langle\langle \text{MATH_I_162} \rangle\rangle$ is a conclusion of the rescaled integral and not an additional weighting rule.

At an isolated fibre point $\langle\langle \text{MATH_I_163} \rangle\rangle$, $\langle\langle \text{MATH_I_164} \rangle\rangle$, the derivative instead has rank two and $\langle\langle \text{MATH_D_26} \rangle\rangle$. Under the F-side coercivity condition this full-rank local geometry yields a two-dimensional normal Gaussian integral, hence evidence of order $\langle\langle \text{MATH_I_165} \rangle\rangle$. The sheet has only one normal direction and has evidence of order $\langle\langle \text{MATH_I_166} \rangle\rangle$. This difference of normal dimension is the selection mechanism: after common normalization, the entire isolated family can survive only if its countable accumulation defeats the extra factor of $\langle\langle \text{MATH_I_167} \rangle\rangle$. The Gaussian packing estimate rules that out in the present class.

∴ remark **Remark 4** (A scale comparison, not a pointwise comparison). *It would be incorrect to decide selection by comparing the energy at points of the two fibre components, since both energies vanish. The relevant quantities are the integrals over physical normal neighbourhoods. The sheet-tube substitution has one Jacobian factor $\langle\langle \text{MATH_I_168} \rangle\rangle$, whereas an isolated cell has two. The proof below additionally shows that these local scales can be summed over the entire noncompact fibre without a hidden overlap or enumeration convention. ∴*

For later reference, set $\ll\langle\text{MATH_D_27}\rangle$ The first constant is derived in Lemma [5](#lem:periodic-geometry){reference-type="ref" reference="lem:periodic-geometry"}; the second is strictly positive because its integrand is strictly positive. They separate the global countable geometry from the selected sheet contribution.

::: {#cor:quantitative-allocation .corollary} **Corollary 1** (Quantitative allocation after normalization). *For sufficiently small $\ll\langle\text{MATH_I_169}\rangle$, there are constants $\ll\langle\text{MATH_I_170}\rangle$, depending only on $\ll\langle\text{MATH_I_171}\rangle$, such that $\ll\langle\text{MATH_D_28}\rangle$ and $\ll\langle\text{MATH_D_29}\rangle$*
:::

::: proof *Proof*. The first assertion follows from the positive limit in Proposition [3](#prop:posterior-assembly){reference-type="ref" reference="prop:posterior-assembly"}. The root-cell estimate and the definition of $\ll\langle\text{MATH_I_172}\rangle$ give unnormalised mass at most $\ll\langle\text{MATH_I_173}\rangle$; division by $\ll\langle\text{MATH_I_174}\rangle$ gives the second assertion. The exterior estimate is handled in exactly the same way. This corollary records a posterior-level result; it is not used to prove the earlier unnormalised estimates. :::

8 Proofs of the intermediate results and theorem assembly

The F-side separation, coercivity and exterior-gap hypotheses have already supplied the constants and literal physical allocation in Lemma [5](#lem:periodic-geometry){reference-type="ref" reference="lem:periodic-geometry"}; the Gaussian root sum in that lemma is derived by packing, not assumed.

::: proof *Proof of Lemma [2](#lem:sheet-scaling){reference-type="ref" reference="lem:sheet-scaling"}*. On $\ll\langle\text{MATH_I_175}\rangle$ use $\ll\langle\text{MATH_I_176}\rangle$, $\ll\langle\text{MATH_I_177}\rangle$. The exact change of variables gives, for a bounded continuous $\ll\langle\text{MATH_I_178}\rangle$, $\ll\langle\text{MATH_D_30}\rangle$ The integrand converges pointwise to $\ll\langle\text{MATH_D_31}\rangle$ and, after multiplying by $\ll\langle\text{MATH_I_179}\rangle$, is dominated by $\ll\langle\text{MATH_I_180}\rangle$. Dominated convergence therefore yields $\ll\langle\text{MATH_D_32}\rangle$ For clarity, the domination is uniform in the moving domain: on $\ll\langle\text{MATH_I_181}\rangle$ one has $\ll\langle\text{MATH_I_182}\rangle$ and $\ll\langle\text{MATH_I_183}\rangle$, while outside that domain the integrand is defined to be zero. The displayed majorant is integrable on $\ll\langle\text{MATH_I_184}\rangle$. Thus (4) is an ambient Lebesgue-integral limit, rather than a formal integration along the sheet. :::

::: proof *Proof of Lemma [3](#lem:root-cells){reference-type="ref" reference="lem:root-cells"}*. For a root $\ll\langle\text{MATH_I_185}\rangle$, write $\ll\langle\text{MATH_I_186}\rangle$. Put $\ll\langle\text{MATH_I_187}\rangle$. The second displayed energy identity gives $\ll\langle\text{MATH_D_33}\rangle$ Moreover $\ll\langle\text{MATH_I_188}\rangle$. For every $\ll\langle\text{MATH_I_189}\rangle$, the resulting Gaussian integral on $\ll\langle\text{MATH_I_190}\rangle$ is bounded by a constant $\ll\langle\text{MATH_I_191}\rangle$, independent of $\ll\langle\text{MATH_I_192}\rangle$ and $\ll\langle\text{MATH_I_193}\rangle$, times $\ll\langle\text{MATH_I_194}\rangle$: $\ll\langle\text{MATH_D_34}\rangle$ Indeed, $\ll\langle\text{MATH_I_195}\rangle$ and $\ll\langle\text{MATH_I_196}\rangle$ give $\ll\langle\text{MATH_D_35}\rangle$ Thus one may take $\ll\langle\text{MATH_I_197}\rangle$, which is independent of the physical root $\ll\langle\text{MATH_I_198}\rangle$ and of $\ll\langle\text{MATH_I_199}\rangle$. Lemma [5](#lem:periodic-geometry){reference-type="ref" reference="lem:periodic-geometry"} gives $\ll\langle\text{MATH_I_200}\rangle$, and the full isolated contribution is $\ll\langle\text{MATH_I_201}\rangle$. :::

::: proof *Proof of Lemma [4](#lem:exterior-suppression){reference-type="ref" reference="lem:exterior-suppression"}*. Since $\ll\langle\text{MATH_I_202}\rangle$, Lemma [5](#lem:periodic-geometry){reference-type="ref" reference="lem:periodic-geometry"} applies on the remaining physical complement and gives $\ll\langle\text{MATH_D_36}\rangle$ The same bounds hold after multiplication by a bounded $\ll\langle\text{MATH_I_203}\rangle$. :::

::: proof *Proof of Proposition [3](#prop:posterior-assembly){reference-type="ref" reference="prop:posterior-assembly"}* and *Theorem [1](#thm:periodic-selection){reference-type="ref" reference="thm:periodic-selection"}*. Taking $\ll\langle\text{MATH_I_204}\rangle$ in (4)–(6) gives $\ll\langle\text{MATH_D_37}\rangle$ Divide the corresponding numerator limit by (7). This proves weak convergence against every bounded contin-

uous «<MATH_I_205>»; the bounded-Lipschitz metric metrizes weak convergence on the Polish space «<MATH_I_206>» (the same estimates also give uniform tightness: after truncating «<MATH_I_207>» and retaining finitely many «<MATH_I_208>», the complement is uniformly controlled by Gaussian-tail bounds or an exponentially small physical-complement bound). Finally (5), summed over «<MATH_I_209>» and divided by the positive linear lower bound from (7), gives the asserted «<MATH_I_210>» isolated posterior mass.

To spell out the tightness step, fix «<MATH_I_211>». First choose «<MATH_I_212>» so that the Gaussian tail in the sheet integral outside «<MATH_I_213>» is below a fixed multiple of «<MATH_I_214>». The packing bound then leaves only finitely many cells «<MATH_I_215>» with «<MATH_I_216>»; the remaining cell contribution is bounded by the tail of «<MATH_I_217>» times «<MATH_I_218>». After division by the positive linear evidence lower bound in (7), this tail is uniformly small for small «<MATH_I_219>». The physical complement is uniformly small by (6). Enlarging a compact rectangle to contain the finite sheet segment and the finitely many remaining cells gives a compact set of posterior mass at least «<MATH_I_220>». This proves the proposition, and its convergence and isolated-mass conclusions are exactly those of Theorem [1](#thm:periodic-selection){reference-type="ref" reference="thm:periodic-selection"}. ∴

9 Noncompact assembly: tails, tests, and constants

This section makes explicit the two places at which noncompactness enters the proof. They are sometimes compressed into the phrase "apply Laplace's method and normalize", but that compression would be insufficient here: there is both an unbounded selected sheet and a countable collection of unselected cells. The required controls come from different sources.

First consider the sheet tail. The same rescaling used in the proof of Lemma [2](#lem:sheet-scaling){reference-type="ref" reference="lem:sheet-scaling"} gives, for every «<MATH_I_221>», «<MATH_D_38>» Indeed, extend the rescaled window by zero outside «<MATH_I_222>» and use the same normal majorant as before. This is a uniform estimate rather than a consequence of pointwise convergence. It shows that the physical «<MATH_I_223>»-tail of the selected sheet is controlled directly by the Gaussian prior, independently of the behaviour of «<MATH_I_224>» at infinity.

The second control is the tail of the root family. Let «<MATH_D_39>» The packing argument proves «<MATH_I_225>» as «<MATH_I_226>», and the root-cell estimate gives «<MATH_D_40>» These estimates are genuinely different. The preceding sheet-tail bound is an ambient Gaussian tail on a continuum; the root-tail bound is a Gaussian-weighted packing estimate on a discrete family. Neither can replace the other.

∴ {#prop:compact-containment .proposition} **Proposition 5** (Explicit compact containment). *For every «<MATH_I_227>» there are «<MATH_I_228>» and «<MATH_I_229>» such that the compact set «<MATH_D_41>» satisfies «<MATH_D_42>» ∴*

∴ proof *Proof.* By the positive limit for «<MATH_I_230>», decrease «<MATH_I_231>» if necessary so that «<MATH_D_43>» Choose «<MATH_I_232>» so that the preceding sheet-tail bound, divided by «<MATH_I_233>», is below «<MATH_I_234>». Next choose «<MATH_I_235>» with «<MATH_I_236>». The finitely many retained cells are contained in a compact rectangle because uniform separation makes «<MATH_I_237>» finite. The root-tail bound together with the displayed denominator lower bound controls the posterior mass of all other root cells by «<MATH_I_238>». Finally the exterior estimate divided by that denominator lower bound tends to zero, so a further decrease of «<MATH_I_239>» bounds the remainder by «<MATH_I_240>». The three regions cover «<MATH_I_241>», proving the claim. ∴

The order in this proof matters. One first chooses spatial truncations $\llangle \text{MATH_I_242} \rrangle$ from quantities which do not depend on $\llangle \text{MATH_I_243} \rrangle$, and only then takes the small-noise threshold. Reversing that order would permit an $\llangle \text{MATH_I_244} \rrangle$ -dependent compact set and would not prove tightness. This explicit order is also why a summable envelope cannot simply be inserted as a hidden assumption: the envelope must be derived in a form that is uniform in the relevant noise range.

For completeness, let $\llangle \text{MATH_I_245} \rrangle$ be bounded and continuous. Split its unnormalised numerator into the sheet tube, the root cells, and the physical complement. Lemma [2](#lem:sheet-scaling){reference-type="ref" reference="lem:sheet-scaling"} supplies the first-order sheet limit; the other two pieces are bounded by $\llangle \text{MATH_I_246} \rrangle$ times Lemmas [3](#lem:root-cells){reference-type="ref" reference="lem:root-cells"} and [4](#lem:exterior-suppression){reference-type="ref" reference="lem:exterior-suppression"}. Therefore $\llangle \text{MATH_D_44} \rrangle$ The same formula with $\llangle \text{MATH_I_247} \rrangle$ gives the strictly positive denominator limit. Taking their quotient identifies the limiting functional with $\llangle \text{MATH_I_248} \rrangle$. Since the ambient state space is Polish, convergence against every bounded continuous test is weak convergence, and hence convergence in the bounded-Lipschitz metric. Proposition [5](#prop:compact-containment){reference-type="ref" reference="prop:compact-containment"} is a separate quantitative certificate of the required noncompact tail control; it is included to make the probability-law step auditable without relying on an unnamed compactness theorem.

10 Sharpness and why local regularity is insufficient

The theorem uses three profile conditions because they exclude three distinct ways in which a local calculation can fail to globalize. We discuss these global roles in turn. First, uniform separation does not merely make root labels convenient: it converts the root set into a uniformly locally finite physical set and produces the Gaussian packing sum. If roots are allowed to accumulate in a bounded interval, a family of disjoint root-centred cells with a common radius is impossible; if their number grows too quickly in the tails, the root-cell contributions need not be summable under a specified prior.

Second, uniform local coercivity is equally indispensable. A simple zero at each individual root supplies a root-dependent linear lower bound, but the proof requires one constant $\llangle \text{MATH_I_249} \rrangle$ for all cells. Without it, a sequence of flatter and flatter roots can have cell widths which compensate for the nominal codimension-two penalty. The escaping residual in Proposition [6](#prop:escaping-sharpness){reference-type="ref" reference="prop:escaping-sharpness"} is a concrete warning: its integer zeros remain separated and regular, but its slopes decay with the root index, so its normalized posterior can leave every compact set along a suitable noise sequence.

Third, the exterior gap is a residual statement, not a posterior-tail statement. It says that after the physically declared sheet strip and root cells have been removed, no low-energy channel remains. This is what turns the complement into the exponential term in Lemma [4](#lem:exterior-suppression){reference-type="ref" reference="lem:exterior-suppression"}. A weak condition such as "the zero set is closed" would not prevent a nonzero residual from becoming arbitrarily small far from the fibre, and could leave an additional competing near-minimizer family invisible to a local fibre description.

These boundaries also explain the scope of the result. The theorem is not a claim for arbitrary noncompact or stratified fibres, arbitrary priors, or arbitrary dimensions. It is a complete posterior theorem for a broad explicit residual class in which the physical allocation and all global tails can be derived. The sharpness example shows that this is not a cosmetic restriction, while Propo-

sition [2](#prop:aperiodic-family){reference-type="ref" reference="prop:aperiodic-family"} shows that it is not restricted to periodic models or to a single trigonometric witness.

11 Derivation of the selected coefficient and a reusable proof pattern

It is useful to isolate the calculation behind the coefficient in the limit, both as a check on constants and as a template for related mixed-codimension models. In the sheet tube, the exact normal substitution $\ll\langle\text{MATH_I_250}\rangle$ gives $\ll\langle\text{MATH_D_45}\rangle$ For a fixed $\ll\langle\text{MATH_I_251}\rangle$, the small-noise normal energy therefore converges to $\ll\langle\text{MATH_I_252}\rangle$. The Gaussian integral in that normal coordinate is $\ll\langle\text{MATH_D_46}\rangle$ The two-dimensional prior contributes $\ll\langle\text{MATH_I_253}\rangle$. After the normal integration the coefficient in (4) is therefore $\ll\langle\text{MATH_D_47}\rangle$ Its constant factor cancels under posterior normalization, leaving the selected marginal density $\ll\langle\text{MATH_I_254}\rangle$.

The same calculation can be expressed geometrically. The sheet tangent is the $\ll\langle\text{MATH_I_255}\rangle$ -direction, and the image of a unit normal under $\ll\langle\text{MATH_I_256}\rangle$ has norm $\ll\langle\text{MATH_I_257}\rangle$. Thus the coefficient is the ordinary Gaussian normal mass divided by the normal Jacobian. The direct rescaling proof is still needed: it justifies this geometric reading in an ambient noncompact posterior, fixes the prefactor, and proves a domination estimate that is uniform in $\ll\langle\text{MATH_I_258}\rangle$.

The proof has a reusable sequence of obligations for a residual with several candidate components:

1. identify a literal partition of the physical source space into selected tubes, competing cells, and a complement;
1. compute the normal rescaling and an integrable domination on each selected component;
1. obtain a uniform normal lower bound in every competing cell and derive a summable prior-weighted cell bound;
1. prove a positive energy gap on the complement; and
1. compare the resulting evidence orders before dividing by the actual common normalization.

The present theorem realizes all five steps without introducing a separate atlas measure. In particular, the countable sum occurs only after the physical root cells have been fixed, and its summand is dictated by the prior and the root-centred coordinate estimate.

There are two common but invalid shortcuts that this order avoids. The first is to calculate a local coarea weight on each stratum and then silently add the answers over an infinite family. Such a calculation provides no summability or tightness statement. The second is to normalize a selected component before verifying that the rest of the physical space is negligible. That produces a surrogate conditional law, not necessarily the limit of the ambient posterior. The sheet-tail bound, root-tail bound, and exterior estimate show why the two shortcuts happen to be valid in this model only after the required global estimates have been established.

The same proof pattern also clarifies what would have to change in higher dimension. A selected stratum of codimension $\ll\langle\text{MATH_I_259}\rangle$ normally produces an evidence factor $\ll\langle\text{MATH_I_260}\rangle$ after a $\ll\langle\text{MATH_I_261}\rangle$ -dimensional normal substitution. A competing stratum of larger codimension is suppressed only when its local coefficients can be summed and the complement has no comparable low-energy region. This statement is a guide rather than a claimed general theorem: the present paper proves it only for the explicit planar residual, where the normal forms, partition, and estimates are all available directly.

12 Detailed verification of the sharpness mechanism

The escaping example is included to separate the theorem's global assumptions from the weaker assertion that every root is locally regular. At the integer root $\ll\langle\text{MATH_I_262}\rangle$ of $\ll\langle\text{MATH_D_48}\rangle$ one has $\ll\langle\text{MATH_I_263}\rangle$. The standard local Laplace coefficient for the one-dimensional Gaussian prior is therefore $\ll\langle\text{MATH_D_49}\rangle$ which does not decay with $\ll\langle\text{MATH_I_264}\rangle$. This cancellation is the mechanism behind escape: the prior penalizes distant roots, but their increasingly small local slope cancels that penalty at the local evidence scale.

For $\ll\langle\text{MATH_I_265}\rangle$, the window radius $\ll\langle\text{MATH_I_266}\rangle$ is at most a constant multiple of $\ll\langle\text{MATH_I_267}\rangle$ for $\ll\langle\text{MATH_I_268}\rangle$. In particular, the windows used in Proposition [6](#prop:escaping-sharpness){reference-type="ref" reference="prop:escaping-sharpness"} can be chosen disjoint. On each such window Taylor's theorem keeps $\ll\langle\text{MATH_I_269}\rangle$ at noise scale while the Gaussian prior changes by a uniformly bounded factor: the relative displacement from $\ll\langle\text{MATH_I_270}\rangle$ is controlled by $\ll\langle\text{MATH_I_271}\rangle$. Hence every one of the $\ll\langle\text{MATH_I_272}\rangle$ windows contributes a common multiple of $\ll\langle\text{MATH_I_273}\rangle$ to the unnormalised evidence. In contrast, a fixed compact set intersects only finitely many regular-root neighbourhoods, and its total evidence is at most a fixed multiple of $\ll\langle\text{MATH_I_274}\rangle$.

After normalization, the fixed compact set has mass $\ll\langle\text{MATH_I_275}\rangle$. This is a posterior-level failure, not merely a failure of a formal series: the measures are actual normalized finite-noise Gaussian residual posteriors. It also pinpoints the failed F-side assumptions. Their roots remain uniformly separated, but neither a common local coercivity constant nor a positive exterior residual gap can persist because the amplitude $\ll\langle\text{MATH_I_276}\rangle$ tends to zero. The example therefore does not contradict Theorem [1](#thm:periodic-selection){reference-type="ref" reference="thm:periodic-selection"}; it demonstrates why the theorem cannot be deduced from local regularity plus separation alone.

13 Sharpness and relation to existing results

Supplementary material contains Lean 4 formalizations of auxiliary calibrations (including a variable-Jacobian example and radial-complexity pairs). They are not used in the proof of Theorem [1](#thm:periodic-selection){reference-type="ref" reference="thm:periodic-selection"}.

The following proposition identifies the global obstruction that makes the uniform residual controls in the theorem necessary.

∴ {#prop:escaping-sharpness .proposition} **Proposition 6** (Local regularity does not imply posterior tightness). *Let $\ll\langle\text{MATH_I_277}\rangle$ and, for $\ll\langle\text{MATH_I_278}\rangle$, set $\ll\langle\text{MATH_I_279}\rangle$. For $\ll\langle\text{MATH_I_280}\rangle$, set $\ll\langle\text{MATH_D_50}\rangle$ For a measurable set $\ll\langle\text{MATH_I_281}\rangle$, define $\ll\langle\text{MATH_D_51}\rangle$ Then $\ll\langle\text{MATH_I_282}\rangle$ is the normalized Gaussian residual posterior on $\ll\langle\text{MATH_I_283}\rangle$ with residual $\ll\langle\text{MATH_I_284}\rangle$ and noise $\ll\langle\text{MATH_I_285}\rangle$. The function $\ll\langle\text{MATH_I_286}\rangle$ is smooth, every $\ll\langle\text{MATH_I_287}\rangle$ is a regular zero, and $\ll\langle\text{MATH_D_52}\rangle$ Consequently $\ll\langle\text{MATH_I_288}\rangle$ is not tight and cannot have a bounded-Lipschitz probability-law limit along $\ll\langle\text{MATH_I_289}\rangle$. ∴*

∴ proof *Proof.* The derivative formula gives $\ll\langle\text{MATH_I_290}\rangle$. Thus each integer root is locally regular, while its local Gaussian coefficient is independent of $\ll\langle\text{MATH_I_291}\rangle$: $\ll\langle\text{MATH_D_53}\rangle$ Fix $\ll\langle\text{MATH_I_292}\rangle$ small and, for $\ll\langle\text{MATH_I_293}\rangle$, use the disjoint windows $\ll\langle\text{MATH_I_294}\rangle$. Taylor's theorem gives $\ll\langle\text{MATH_I_295}\rangle$ on those windows for all sufficiently large $\ll\langle\text{MATH_I_296}\rangle$; the Gaussian prior changes by a bounded factor there because $\ll\langle\text{MATH_I_297}\rangle$. In the unnormalised convention used in this paper, every window contributes at least $\ll\langle\text{MATH_I_298}\rangle$, so

«<MATH_I_299>». On the other hand, for a fixed compact «<MATH_I_300>», a finite nearest-integer cell cover and the lower slope of «<MATH_I_301>» on each cell give «<MATH_D_54>» Thus «<MATH_I_302>». The Lean supplement uses the normalization-equivalent «<MATH_I_303>» likelihood: it therefore checks the same normalized posterior and its non-tightness, while its un-normalised window and evidence bounds differ from those displayed here by one common positive «<MATH_I_304>»-dependent factor. The example lies outside the uniformly-separated profile class, so it is a sharpness boundary rather than a false counterexample to Theorem [1](#thm:periodic-selection){reference-type="ref" reference="thm:periodic-selection"}. ::

The contribution is a complete Gaussian-posterior selection theorem for a periodicity-free uniformly separated profile class with mixed normal codimensions. Stuart gives the Bayesian inverse-problem baseline [Stuart, 2010]; Léger–Vialard give a regular geometric-Laplace benchmark [Léger–Vialard, 2023]; and Minea–Păltănea show that countably many extrema require a genuine tail analysis [Minea–Păltănea, 2015]. Recent residual-posterior work establishes determinant-corrected limits for a unique smooth graph with nonsingular state Jacobian [Cottom–Olsson, 2026]; CoCoS treats the regular full-row-rank coarea/Fixman limit [Xu et al., 2026]; and manifold lifting studies the regular constrained posterior geometry [Au–Graham–Thiery, 2023]. Ievlev’s recent stratified theory uses a compact finite stratification and a supplied adapted normal-cell disintegration [Ievlev, 2026].

Unlike those regular, unique-graph, or compact-finite frameworks, Theorem [1](#thm:periodic-selection){reference-type="ref" reference="thm:periodic-selection"} treats a fibre with a noncompact sheet, a countably infinite competing family, and two normal codimensions. Ievlev’s compact base cannot accommodate the noncompact sheet here, while the unique-graph assumption of Cottom–Olsson fails because the present fibre has mixed normal codimensions. The physical allocation, Gaussian-tail control, and exterior gap are all derived from verifiable F-side separation and residual geometry. The result is established for this explicit, periodicity-free model class; it is not presented as a theorem for every noncompact stratified posterior.

14 Analytic position and limits of transfer

The result lies at the intersection of three familiar lines of analysis, but none alone supplies the displayed theorem. Bayesian inverse-problem theory provides the finite-noise posterior and the question of its small-noise behaviour [Stuart, 2010]. Geometric Laplace and coarea calculations explain the local normal-Jacobian factor at a regular component [Léger–Vialard, 2023; Au–Graham–Thiery, 2023]. Finally, results on countably many extrema show that infinite local contributions require a separate global tail argument [Minea–Păltănea, 2015]. The present model needs all three elements at once because the selected component is noncompact, while the competing components are countable and have a different normal dimension.

For a compact regular constraint manifold, a finite atlas and a uniform normal neighbourhood are often available from compactness. In that setting local coarea coefficients can be patched by finitely many charts. Here the sheet is noncompact and the other component has infinitely many physical pieces, so compactness cannot supply either the root-cell summation or the tail needed for a probability law. The standard Gaussian prior is not a decorative choice: its «<MATH_I_305>»-tail is used twice, once for the continuous sheet tail and once as the derived weight in the discrete packing estimate.

The theorem should also be distinguished from a result about one unique graph of constraint solutions. A unique-graph hypothesis produces one local normal structure, whereas «<MATH_I_306>» contains both a sheet and isolated points. The two components cannot be combined by a single

fixed-rank Jacobian determinant. Their relative contribution is determined only after their normal Gaussian integrals are compared at the evidence scale. This is why the conclusion is a selection law rather than merely a formula for an unnormalized local coefficient.

Recent stratified approaches are valuable comparators, but the theorem makes no claim to subsume them. A compact finite stratification with a supplied adapted normal-cell disintegration begins from global structural information of a different type [Ievlev, 2026]. In the present class, the physical allocation, root summability, and exterior estimate are derived directly from the scalar residual profile. Conversely, the argument here does not provide a general measurable normal atlas, a coarea disintegration for arbitrary nonlinear maps, or a theorem for variable-dimensional strata in arbitrary ambient dimension.

The precise transfer lesson is therefore modest but useful. If a new residual admits (i) an exact sheet-tube rescaling with an integrable majorant, (ii) physical competing cells with a uniform normal lower bound, (iii) a prior-weighted sum of their local coefficients, and (iv) a literal low-energy exclusion on the remaining complement, then the architecture of Sections [7](#sec:proof){reference-type="ref" reference="sec:proof"}–[8](#sec:assembly-details){reference-type="ref" reference="sec:assembly-details"} can be reused. The verification of these four inputs is model-specific. It cannot be replaced by saying that all roots are "regular", or that a formal atlas has been named.

14.1 Why the posterior is an ambient law

Another boundary concerns the object that converges. The limiting measure $\llcorner \text{MATH_I_307} \gg$ is supported on a one-dimensional set, while every $\llcorner \text{MATH_I_308} \gg$ is an absolutely continuous probability measure on $\llcorner \text{MATH_I_309} \gg$. The convergence is therefore not equality of densities on a fixed reference measure; it is weak convergence of ambient probability measures. The rescaling in Lemma [2](#lem:sheet-scaling){reference-type="ref" reference="lem:sheet-scaling"} retains the ambient normal variable long enough to prove the limit against $\llcorner \text{MATH_I_310} \gg$. Only after integration and normalization does the limit become a measure carried by the sheet.

This distinction prevents two further ambiguities. First, a change in a symbolic branch allocation is irrelevant unless it changes the actual ambient integral. Second, an unnormalised asymptotic coefficient is not itself a posterior claim: the competing evidence and the denominator must be controlled on the same noise scale. The explicit use of $\llcorner \text{MATH_I_311} \gg$ and the compact-containment proposition ensure that both points are resolved in the theorem rather than delegated to informal Laplace terminology.

15 Formal verification and reproducibility

The Lean development is deliberately aligned with, but does not overstate, the analytic proof above. Its present coverage is as follows.

::: description P1UniformlySeparatedCleanFibreClass.lean proves the literal fibre and energy identities, common physical geometry, local isolated coercivity, and the residual-side exterior gap used in Lemma [5](#lem:periodic-geometry){reference-type="ref" reference="lem:periodic-geometry"}.

The P2 uniformly-separated rescaling development proves the exact normal rescaling identity and its profile-independent exponent bound. The companion P2 periodic-sheet-law development additionally formalizes the dominated sheet-integral limit and ambient-observable limit for the periodic admissible subclass. The periodicity-free dominated-convergence step in Lemma [2](#lem:sheet-scaling){reference-type="ref" reference="lem:sheet-scaling"} remains the analytic argument written here.

The root packing sum, its uniform Gaussian cell integration, and the conversion of the literal gap into the integrated estimate of Lemmas [3](#lem:root-cells){reference-type="ref" reference="lem:root-cells"}–[4](#lem:exterior-suppression){reference-type="ref" reference="lem:exterior-suppression"} are analytic parts of this manuscript; Lean supplies their geometric inputs, not a claimed countable-atlas theorem.

Normalization, uniform tightness, and bounded-Lipschitz convergence in Proposition [3](#prop:posterior-assembly){reference-type="ref" reference="prop:posterior-assembly"} are analytic consequences of the four displayed estimates. No Lean theorem is cited as a substitute for this assembly.

`P3PeriodicProfileSharpness.lean` formalizes the normalized escaping posterior and its failure of tightness. :::

Thus the formal files certify exact algebraic and selected measure-theoretic components. The paper independently supplies the noncompact countable-family estimates and probability-law argument. Exact commands, module-to-claim links, and trusted-boundary checks are collected in the reproducibility supplement.

15.1 References

1. A. M. Stuart, *Inverse problems: a Bayesian perspective*, Acta Numerica **19** (2010), 451–559. <<https://doi.org/10.1017/S0962492910000061>>
1. F. Léger and F.-X. Vialard, *A geometric Laplace method*, Pure and Applied Analysis **5** (2023), no. 4, 1041–1080.
1. B. Minea and E. Păltănea, **Laplace’s method for the case of a countably infinite set of global maximum points**, Bulletin mathématique de la Société des Sciences Mathématiques de Roumanie **58(106)** (2015), no. 1, 81–94.
1. P. Ievlev, *Laplace asymptotics near stratified minimum sets*, arXiv:2608.02626 (2026). <<https://arxiv.org/abs/2608.02626>>
1. K. X. Au, M. M. Graham, and A. H. Thiery, **Manifold lifting: scaling Markov chain Monte Carlo to the vanishing noise regime**, Journal of the Royal Statistical Society Series B **85** (2023), 757–784.
1. J. Xu, D. Zeng, J. Paisley, and Q. Zhao, **The right measure for physics-constrained generation: a co-area correction for posterior-consistent PDE inverse problems**, arXiv:2606.04804 (2026).
1. J. Cottom and E. Olsson, **Constraint residuals, graph posteriors, and determinant-corrected full-space targets in Bayesian inverse problems**, arXiv:2606.09594 (2026). <<https://arxiv.org/abs/2606.09594>>
1. M. Dashti and A. M. Stuart, *The Bayesian approach to inverse problems*, in R. Ghanem, D. Higdon, and H. Owhadi (eds.), **Handbook of Uncertainty Quantification**, Springer, 2017. <<https://arxiv.org/abs/1302.6989>>
1. C. Schillings, B. Sprungk, and P. Wacker, **On the convergence of the Laplace approximation and noise-level-robustness of Laplace-based Monte Carlo methods for Bayesian inverse problems**, Numerische Mathematik **145** (2020), 915–971. <<https://doi.org/10.1007/s00211-020-01131-1>>
1. T. Helin and R. Kretschmann, **Non-asymptotic error estimates for the Laplace approximation in Bayesian inverse problems**, Numerische Mathematik **150** (2022), 521–549. <<https://doi.org/10.1007/s00211-021-01266-9>>

1. T. Bui-Thanh, O. Ghattas, J. Martin, and G. Stadler, *A computational framework for infinite-dimensional Bayesian inverse problems. Part I: The linearized case, with application to global seismic inversion*, SIAM Journal on Scientific Computing **35** (2013), no. 6, A2494–A2523.
1. S. L. Cotter, G. O. Roberts, A. M. Stuart, and D. White, *MCMC methods for functions: modifying old algorithms to make them faster*, Statistical Science **28** (2013), no. 3, 424–446. <<https://doi.org/10.1214/13-STS421>>
1. M. Hairer, A. M. Stuart, and S. J. Vollmer, *Spectral gaps for a Metropolis-Hastings algorithm in infinite dimensions*, Annals of Applied Probability **24** (2014), no. 6, 2455–2490.
1. T. Cui, K. J. H. Law, and Y. M. Marzouk, *Dimension-independent likelihood-informed MCMC*, Journal of Computational Physics **304** (2016), 109–137. <<https://doi.org/10.1016/j.jcp.2015.10.008>>
1. A. Beskos, M. Girolami, S. Lan, P. E. Farrell, and A. M. Stuart, *Geometric MCMC for infinite-dimensional inverse problems*, Journal of Computational Physics **335** (2017), 327–351.
1. P. Diaconis, S. Holmes, and M. Shahshahani, *Sampling from a manifold*, in *Advances in Modern Statistical Theory and Applications: A Festschrift in Honor of Morris L. Eaton*, IMS Collections **10** (2013), 102–125.
1. M. Brubaker, M. Salzmann, and R. Urtasun, *A family of MCMC methods on implicitly defined manifolds*, Proceedings of AISTATS 2012, PMLR **22** (2012), 161–172. <<https://proceedings.mlr.press/>>
1. H. Federer, *Curvature measures*, Transactions of the American Mathematical Society **93** (1959), no. 3, 418–491.
1. L. Simon, *Lectures on Geometric Measure Theory*, Proceedings of the Centre for Mathematical Analysis, Australian National University, vol. 3, 1983.
1. L. C. Evans and R. F. Gariepy, *Measure Theory and Fine Properties of Functions*, CRC Press, 1992; revised edition, 2015.
1. P. Mattila, *Geometry of Sets and Measures in Euclidean Spaces: Fractals and Rectifiability*, Cambridge University Press, 1995.
1. R. Bott, *Nondegenerate critical manifolds*, Annals of Mathematics **60** (1954), no. 2, 248–261. <<https://www.jstor.org/stable/1969631>>
1. S. Watanabe, *Algebraic Geometry and Statistical Learning Theory*, Cambridge University Press, 2009. <<https://doi.org/10.1017/CBO9780511800474>>
1. L. de Moura, S. Kong, J. Avigad, F. van Doorn, and J. von Raumer, *The Lean theorem prover (system description), in Automated Deduction–CADE-25*, Lecture Notes in Computer Science **9195**, Springer, 2015, 378–388.
1. The mathlib Community, *The Lean mathematical library*, Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), 367–381. <<https://doi.org/10.1145/3372885.3373824>>